Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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1. Цели и задачи работы

Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

Задачи лабораторной работы

1 Изучить перенаправление ввода-вывода

2 Изучить работу фильтров

3 Изучить команду поиска

4 Ознакомиться с управлением процессами

5 Ознакомиться с командами df du

2. Процесс выполнения лабораторной работы

Перенаправление ввода-вывода

```
snbondarenko@snbondarenko:~$ ls /etc/ > file.txt
snbondarenko@snbondarenko:~$ ls >> file.txt
snbondarenko@snbondarenko:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
cifs-utils
cockpit
```

Рисунок 1: Запись в файл

Работа фильтра

```
snbondarenko@snbondarenko:~$ grep .conf file.txt > conf.txt
snbondarenko@snbondarenko:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
```

Рисунок 2: Поиск расширения .conf

Команда поиска

```
/home/snbondarenko/.npm/_cacache/index-v5/f3/6c/c3986891a2eed646eaa7d914d91cffc8d6421272c959bdb5a87486645
945
/home/snbondarenko/.npm/_cacache/index-v5/84/ce
/home/snbondarenko/.npm/_cacache/index-v5/09/c0
/home/snbondarenko/.npm/_cacache/index-v5/09/c0/cce94e2c664a8f390a8f1ade3346d5c95f3ced5cf6fc29def30220242
c53
/home/snbondarenko/.npm/_cacache/index-v5/ce
/home/snbondarenko/.npm/_cacache/index-v5/59/e7/cb225f160e685933e8439d3cf624ef721a58996cfad55fb2000f59e2e
dcc
/home/snbondarenko/.npm/_cacache/index-v5/cd
/home/snbondarenko/.npm/_cacache/index-v5/c9
/home/snbondarenko/.npm/_cacache/index-v5/c0
/home/snbondarenko/.npm/_cacache/index-v5/9f/ae/c077854801f7b4b5cf7eea2bb68ca2bc53f0b53cea310cda0b203217f
693
/home/snbondarenko/.npm/_cacache/index-v5/44/44/c9a2c5dc2999cbe868c370ef479e17dfd1e86117d1350482a1d990504
757
/home/snbondarenko/.npm/_cacache/index-v5/af/cf
/home/snbondarenko/.npm/_cacache/index-v5/c3
/home/snbondarenko/.npm/_cacache/index-v5/a1/5e/c9aa1c46bf0c67511379e97bd07fca87532422f2506214f94db886116
0d3
/home/snbondarenko/.npm/_cacache/index-v5/78/34/c1e131014ad3a4ee13f096e70f74c03e68837bc2c8188a44d554f3f3b
a2e
/home/snbondarenko/.npm/_cacache/index-v5/ca
/home/snbondarenko/snap/hugo/common
/home/snbondarenko/snap/hugo/current
/home/snbondarenko/conf.txt
snbondarenko@snbondarenko:~$
```

Рисунок 3: Поиск файлов

Команда поиска

```
/etc/foomatic/hashes.d/hashes.new
find: ‘/etc/libvirt’: Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
/etc/logrotate.d/httpd
find: ‘/etc/lvm/archive’: Permission denied
find: ‘/etc/lvm/backup’: Permission denied
find: ‘/etc/lvm/cache’: Permission denied
find: ‘/etc/lvm/devices’: Permission denied
find: ‘/etc/nftables’: Permission denied
find: ‘/etc/openvpn/client’: Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: ‘/etc/openvpn/server’: Permission denied
find: ‘/etc/pki/rsyslog’: Permission denied
find: ‘/etc/polkit-1/localauthority’: Permission denied
find: ‘/etc/polkit-1/rules.d’: Permission denied
find: ‘/etc/sos/cleaner’: Permission denied
find: ‘/etc/ssh/sshd_config.d’: Permission denied
find: ‘/etc/sssd’: Permission denied
find: ‘/etc/sudoers.d’: Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
find: ‘/etc/userdb’: Permission denied
/etc/sysconfig/htcacheclean
/etc/systemd/system/httpd.service.d
/etc/systemd/system/multi-user.target.wants/happd.service
```

Рисунок 4: Поиск файлов

Управление процессами

```
snbondarenko@snbondarenko:~$
snbondarenko@snbondarenko:~$ find ~ -name "log*" > logfile &
[1] 6385
snbondarenko@snbondarenko:~$
[1]+  Done                    find ~ -name "log*" > logfile
snbondarenko@snbondarenko:~$ rm logfile
snbondarenko@snbondarenko:~$
```

Рисунок 5: Фоновый запуск процесса

Управление процессами

```
snbondarenko@snbondarenko:~$
snbondarenko@snbondarenko:~$ gedit &
[1] 6424
snbondarenko@snbondarenko:~$
snbondarenko@snbondarenko:~$ ps | grep gedit
   6424 pts/0    00:00:00 gedit
snbondarenko@snbondarenko:~$
snbondarenko@snbondarenko:~$ kill 6424
[1]+  Terminated              gedit
snbondarenko@snbondarenko:~$
snbondarenko@snbondarenko:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

Команды df и du

```
DF(1)                                User Commands                                DF(1)

NAME
       df - report file system space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This  manual  page documents the GNU version of df.  df displays the amount of space available on
       the file system containing each file name argument.  If no file name is given, the  space  avail-
       able on all currently mounted file systems is shown.  Space is shown in 1K blocks by default, un-
       less the environment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

       If  an  argument  is the absolute file name of a device node containing a mounted file system, df
       shows the space available on that file system rather than on the file system containing  the  de-
       vice node.  This version of df cannot show the space available on unmounted file systems, because
       on  most kinds of systems doing so requires non-portable intimate knowledge of file system struc-
       tures.

OPTIONS
       Show information about the file system on which each FILE resides, or all  file  systems  by  de-
       fault.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              include pseudo, duplicate, inaccessible file systems

       -B, --block-size=SIZE
              scale  sizes  by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576
              bytes; see SIZE format below

       --direct
              show statistics for a file instead of mount point

 Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

Команды df и du

```
DU(1)                                User Commands                                DU(1)

NAME
       du - estimate file space usage

SYNOPSIS
       du [OPTION]... [FILE]...
       du [OPTION]... --files0-from=F

DESCRIPTION
       Summarize device usage of the set of FILEs, recursively for directories.

       Mandatory arguments to long options are mandatory for short options too.

       -0, --null
              end each output line with NUL, not newline

       -a, --all
              write counts for all files, not just directories

       --apparent-size
              print  apparent  sizes  rather  than  device  usage; although the apparent size is usually
              smaller, it may be larger due to holes in ('sparse') files, internal fragmentation,  indi-
              rect blocks, and the like

       -B, --block-size=SIZE
              scale  sizes  by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576
              bytes; see SIZE format below

       -b, --bytes
              equivalent to '--apparent-size --block-size=1'

       -c, --total
              produce a grand total

       -D, --dereference-args
 Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

Команды df и du

```
snbondarenko@snbondarenko:~$ man du
snbondarenko@snbondarenko:~$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 91000500  63575932  59% /
devtmpfs         4011288        0   4011288   0% /dev
tmpfs            4035664       92   4035572   1% /dev/shm
tmpfs            1614268     1952   1612316   1% /run
tmpfs               1024        0      1024   0% /run/credentials/systemd-journald.service
tmpfs            4035664     5728   4029936   1% /tmp
/dev/nvme0n1p3 155186176 91000500  63575932  59% /home
/dev/nvme0n1p2   1992552   518800   1352512  28% /boot
/dev/loop1        115712   115712         0 100% /var/lib/snapd/snap/hugo/25676
/dev/loop0         68480    68480         0 100% /var/lib/snapd/snap/core24/1349
/dev/loop2         49280    49280         0 100% /var/lib/snapd/snap/snapd/25935
tmpfs               1024        0      1024   0% /run/credentials/systemd-resolved.service
tmpfs             807132      164    806968   1% /run/user/1056
/dev/sr0         2677920  2677920         0 100% /run/media/snbondarenko/Fedora-WS-Live-43
snbondarenko@snbondarenko:~$
```

Рисунок 9: Справка по команде du

Команды df и du

```
4         ./.npm/_cacache/index-v5/aa
4         ./.npm/_cacache/index-v5/a7/81
4         ./.npm/_cacache/index-v5/a7
4         ./.npm/_cacache/index-v5/07/67
4         ./.npm/_cacache/index-v5/07
4         ./.npm/_cacache/index-v5/30/1a
4         ./.npm/_cacache/index-v5/30
424       ./.npm/_cacache/index-v5
115644    ./.npm/_cacache
115708    ./.npm
16        ./snap/hugo/25676/.config/go/telemetry/local
0         ./snap/hugo/25676/.config/go/telemetry/upload
16        ./snap/hugo/25676/.config/go/telemetry
16        ./snap/hugo/25676/.config/go
16        ./snap/hugo/25676/.config
16        ./snap/hugo/25676
0         ./snap/hugo/common
20        ./snap/hugo
20        ./snap
0         ./monthly
0         ./reports/monthly/monthly
0         ./reports/monthly
0         ./reports
4         ./ski.plases/equipment
0         ./ski.plases/plans
4         ./ski.plases
0         ./australia
0         ./play/games/play
0         ./play/games
0         ./play
1514764   .
snbondarenko@snbondarenko:~$
```

Рисунок 10: Запуск команды du

Команда поиска

```
/home/snbondarenko/.npm/_cacache/index-v5/52
/home/snbondarenko/.npm/_cacache/index-v5/52/4d
/home/snbondarenko/.npm/_cacache/index-v5/aa
/home/snbondarenko/.npm/_cacache/index-v5/aa/bf
/home/snbondarenko/.npm/_cacache/index-v5/a7
/home/snbondarenko/.npm/_cacache/index-v5/a7/81
/home/snbondarenko/.npm/_cacache/index-v5/07
/home/snbondarenko/.npm/_cacache/index-v5/07/67
/home/snbondarenko/.npm/_cacache/index-v5/30
/home/snbondarenko/.npm/_cacache/index-v5/30/1a
/home/snbondarenko/snap
/home/snbondarenko/snap/hugo
/home/snbondarenko/snap/hugo/25676
/home/snbondarenko/snap/hugo/25676/.config
/home/snbondarenko/snap/hugo/25676/.config/go
/home/snbondarenko/snap/hugo/25676/.config/go/telemetry
/home/snbondarenko/snap/hugo/25676/.config/go/telemetry/local
/home/snbondarenko/snap/hugo/25676/.config/go/telemetry/upload
/home/snbondarenko/snap/hugo/common
/home/snbondarenko/monthly
/home/snbondarenko/reports
/home/snbondarenko/reports/monthly
/home/snbondarenko/reports/monthly/monthly
/home/snbondarenko/ski.plases
/home/snbondarenko/ski.plases/equipment
/home/snbondarenko/ski.plases/plans
/home/snbondarenko/australia
/home/snbondarenko/play
/home/snbondarenko/play/games
/home/snbondarenko/play/games/play
snbondarenko@snbondarenko:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

Вывод

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.